AADI MOHANE

Electronics & Telecommunication Engineering (B.E.)
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PROFESSIONAL SUMMARY

Detail-oriented Third-year Electronics and Telecommunication Engineering student at Shri Sant Gajanan Maharaj College of Engineering (SSGMCE) with professional internship exposure in industrial VLSI testing alongside independent hardware prototyping skills. Experienced in implementing digital logic on physical FPGAs and building microcontroller nodes from scratch. Highly motivated to apply practical hardware debugging and foundational data tools toward Industry 4.0, smart manufacturing, and plant automation projects.

TECHNICAL SKILLS

- **Hardware Prototyping:** 2-Layer PCB Layout (EasyEDA), Circuit Simulation, Through-Hole Soldering, Physical Troubleshooting.
- **Embedded & VLSI:** Xilinx Vivado, Cadence EDA Tools, FPGA Deployment (Boolean Board / Spartan Processor Architecture), Arduino IDE.
- **Engineering Software:** MATLAB (Basic Modeling), Python (Basic Scripting), SQL (Basic Database Queries).
- **Areas of Interest:** Industrial Automation, IoT Hardware Nodes, Smart Manufacturing, Digital Twins.

INDUSTRIAL EXPERIENCE

VLSI Project Trainee May 2026 – June 2026
NIT Rourkela

- Selected for a technical training track designed to bridge the gap between academic theory and real-world semiconductor/VLSI application architectures.
- Worked directly on the digital design, modeling, and physical implementation of computational hardware processing modules.
- Programmed and tested a specialized GCD Processor utilizing Xilinx Vivado to run timing simulations and structural verifications.
- Successfully flashed, deployed, and verified the target logic constraints onto physical Boolean Board FPGAs powered by Spartan processor cores.

Google Student Ambassador September 2025 – February 2026
SSGMCE Campus

- Appointed as the official campus technical representative to handle peer technology awareness campaigns.
- Formulated and executed a high-impact workshop titled 'Career Intelligence: Where AI meets your MBA future' for over 40 stream students.
- Refined team leadership and public presentations vital for presenting engineering updates to plant stakeholders.

INDEPENDENT PERSONAL PROJECTS

Automated Seed Germination Environmental Simulator *IoT/AgriTech Prototype*

- Developed an IoT-enabled microclimate simulator using an Arduino Mega microcontroller to continuously process live sensor feedback for precision agriculture research.
- Integrated DHT11 temperature/humidity sensors and soil moisture sensors with a local LCD interface to deliver real-time environmental data visualization.
- Simulated dynamic microclimate regulation routines controlling relay-driven heating elements and DC exhaust fans to test and evaluate optimal seed viability parameters.

Arduino Nano AC Switch & Home Automation Node *Hardware Design & Prototyping*

- Engineered a compact 2-layer PCB footprint inside EasyEDA capable of safely interfacing and switching high-voltage AC utility loads.
- Wired an isolated relay driver stage to isolate system components and preserve operator safety.
- Embedded standalone infrared sensor headers directly into the board layout to evaluate touchless device control logic.

EDUCATION

Bachelor of Engineering (B.E) in Electronics and Telecommunications 2024 – 2028
Shri Sant Gajanan Maharaj College of Engineering (SSGMCE), Shegaon **CGPA:** 7.70

Higher Secondary Certificate (HSC) **Score:** 87.4%
Aditya Vidhyashram, Pondicherry

Secondary School Certificate (SSC) **Score:** 77.2%
Aditya Vidhyashram, Pondicherry

CERTIFICATIONS & ACHIEVEMENTS

- **VLSI Trainee Certification:** Short-Term Training Program in VLSI Design & FPGA Implementation, NIT Rourkela (June 2026).
- **Google Student Ambassador:** Certificate of Completion & Performance Recognition Vouchers (Feb 2026).
- **IITM Pravartak Python Certification:** IIT Madras Pravartak Technologies Foundation (July 2025).

LANGUAGES

English (Professional), Hindi (Native), Marathi, Tamil (Spoken)